Evaluation and Modelling of Electrically Conductive Polymer Nanocomposites with Carbon Nanotube Networks

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Abstract

Superior electrical, thermal, and mechanical properties of carbon nanotubes (CNTs) have made them effective filler for multifunctional polymer nanocomposites (PNCs). In particular, electrically conductive PNCs filled with CNTs have been researched extensively. These studies aimed to increase the PNCs' electrical conductivity (σ) and to minimize the percolation thresholds (ϕ_c). In this work, we have developed an improved model to describe the CNT networks and thereby evaluate the PNCs' ϕ_c and σ. The new model accounts for the electrical conductance contributed by the continued CNT network across the boundary of adjacent representative volume elements. It more realistically represents the interconnectivity among CNTs and enhances the evaluation of the structure-to-property relationship of PNCs' σ.

Keywords: C. Computational modelling; A. Polymer-matrix composites (PMCs); Nano-structures; B. Electrical properties; Percolation

1. Introduction

Possessing superior electrical, thermal, and mechanical properties, carbon nanotubes (CNTs) [1-5] are considered powerful and effective filler for tailoring the multifunctional properties of polymer nanocomposites (PNCs) [6]. In particular, due to their exceptionally high intrinsic electrical conductivity and high aspect ratio, it is possible to embed only a small amount of CNTs in a polymer matrix to achieve significant enhancement in the PNC's electrical conductivity (σ). Small CNT loadings also allow PNCs to maintain the favorable properties of polymers, including low density, high chemical resistance, and good processability. Accordingly, PNCs filled with CNTs have received much attention for their many potential applications, such as field emission [7], lightning strike protection [8], highly sensitive strain sensors [9-11], and electromagnetic-wave interference materials [12].

The insulator-to-conductor transition of PNCs through the addition of conductive fillers can be modelled by the percolation theory [13]. According to the theory, there exists a critical concentration of CNTs over which further increase in CNT loading would make the PNC become electrically conductive. This critical concentration, namely percolation threshold (ϕ_c), signals the formation of the first conductive path bridging the two terminals of the polymer matrix. This theory is mathematically represented by Equation (1).

$$\sigma = \sigma_0(\phi - \phi_c)^t \text{ for } \quad \phi > \phi_c \qquad (1)$$

where σ_0 is a physical parameter commonly attributed to the intrinsic conductivity of CNTs; and t is the critical exponent, which is known to be dominated by the dimensionality of the system.

The ϕ_c of PNCs filled with CNTs have been reported to range from 0.0025 wt.% [14] to 10.5 wt.% [15]. Upon further increase in CNT loading in a polymer matrix over ϕ_c, PNC's σ will gradually increase and reach a maximum plateau. While extensive research has focused on minimizing ϕ_c [16], maximizing PNC's σ has been another objective in the development of CNT–filled PNCs. To the best of the authors' knowledge, the highest σ reported (i.e., approximately 3000 S/m) was achieved by Kim *at al.* with a PMMA-MWCNT nanocomposite [12].

Extensive experimental studies have shown that PNCs' ϕ_c and σ depend on many factors including the dispersion, alignment and aspect ratio of CNTs. Aguilar *et al.* reported a decrease in PNCs' ϕ_c due to lightly agglomerated dispersion state of CNTs [17]. Variation in the dependence of PNCs' ϕ_c on CNTs' degree of alignment in PNCs with different CNT contents was reported by Du *et al.* [18]. Using CNTs with higher aspect ratios, significant enhancement in PNC's σ was achieved by Yao *et al.* [19].

In addition to experimental studies, fundamental understanding of the conductive percolation mechanism has been established through several theoretical and numerical studies. A number of these studies modelled interconnecting CNTs as random resistor networks embedded in a polymer matrix. The random CNT resistor network model has been progressively developed from two-dimensional (2D) [20] to multi-layered 2D [21] to three-dimensional (3D) [22, 23] scale. In the 3D model, a representative volume element (RVE) is assumed to be periodically repeated to represent the PNC. Hu *et al.* [22] considered the CNTs extruding through the RVE's boundary surface to be part of the adjacent RVE by cutting the extruding portion and relocating it onto the opposite boundary surface. However, such approach overlooks the interconnectivity of CNTs across the boundary surface and requires a RVE of large volume in order to achieve numerical convergence. In this context, Bao *et al.* [23] extended the modelling scheme by considering periodically connective paths on opposite surfaces of RVE. Nevertheless, there are still some possible cases of interconnecting CNTs across the boundary surfaces of adjacent RVEs that are disregarded under this approach. Furthermore, the "cut-and-relocate" approach, which adopted in both studies to handle CNTs extruding out of the RVE, would lead to immediate increase in the number of CNTs in contact with the two terminals, and thereby generated unnecessary bias on the random distribution of CNTs in the polymer matrix.

Along with periodicity, dimensions of the RVE have been identified as a critical factor in ensuring accuracy and efficiency of the simulation. There have been some discussions on the appropriate dimensions of RVE to model the electrical properties of PNCs [23]. In particular, most numerical studies based on 3D models utilize cubic RVEs to capture isotropic percolation of CNTs inside the polymer matrix [22, 23]. However, Shklovskii *et al.* [24] theoretically analyzed two-component systems containing metallic and dielectric regions of isotropic geometry and established that the macroscopic conductivity of such a network could be anisotropic. Percolative conductive systems with high aspect ratio fillers such as CNTs are more likely to be anisotropic. Such concept of anisotropic percolation behaviors for CNT-filled PNCs was experimentally supported by Fu *et al.* [25]. Their study demonstrated the influence of the test sample's thickness, across which σ was measured, on ϕ_c of MWCNT-low density polyethylene (LDPE) nanocomposites. Experimental results showed that the PNC's ϕ_c increased with the sample's thickness. Such observation was rationalized in terms of the probability of CNTs to construct a conductive pathway across opposite surfaces separated by different length scales.

In this study, we have devised a more realistic and efficient modelling tool to account for the interconnectivity of the CNT network across boundary surfaces of adjacent RVEs. In the proposed model, the omission of the “cut-and-relocate” approach, typically used in simulating PNC’s σ and ϕ_c, will eliminate the unnecessary bias caused by the immediate increase in the number of CNTs penetrating the RVE’s two terminals. Furthermore, the concept of anisotropic percolation behavior was further explored by comparing the variation of simulated PNC’s ϕ_c with the change in RVEs’ dimensions in different directions (i.e., both parallel and perpendicular to the direction of electrical current flow). The aforementioned fundamental improvements in the proposed model will facilitate the identification of structure-to-property relationship of electrically conductive CNT. In addition, our simulation results showed that the new model allowed further reduction of the RVE‘s dimensions, and thereby lowered the computation cost of the Monte Carlo simulation without compromising the accuracy of the simulation. By applying the new model to predict PNCs’ ϕ_c and σ, the simulation results were compared to experimental data reported in literature [26-28] and also to results from previous numerical studies [22, 23]. Furthermore, the effect of preferential alignment of CNTs in PNCs with different CNT loadings was investigated using the new modelling scheme.

2. Theoretical Framework

In the simulation model, electrically conductive PNCs can be represented by a cuboid RVE of $L_x \times L_y \times L_z$ embedded with a random distribution of CNTs. A schematic of the cuboid RVE is shown in Figure 1. Electric current is considered to propagate from the left boundary surface (i.e., high voltage electrode) to the right boundary surface (i.e., low voltage electrode).

2.1 Cuboid representative volume element (RVE) of electrically conductive PNCs

Each CNT is considered as a rigid rod and is represented by a line segment starting from ($x_{i,1}$, $y_{i,1}$, $z_{i,1}$) and ending at ($x_{i,2}$, $y_{i,2}$, $z_{i,2}$). In order to model the random distribution of CNTs, the coordinates of the starting and ending points are generated by Equations (2) and (3), respectively.

$$x_{i,1} = L_x \cdot rand \tag{2a}$$

$$y_{i,1} = L_y \cdot rand \tag{2b}$$

$$z_{i,1} = L_z \cdot rand \tag{2c}$$

where *rand* is a uniformly distributed random number in [0,1].

$$x_{i,2} = x_{i,1} + L_{CNT} \cos\varphi_i \sin\theta_i \tag{3a}$$

$$y_{i,2} = y_{i,1} + L_{CNT} \sin\varphi_i \sin\theta_i \tag{3b}$$

$$z_{i,2} = z_{i,1} + L_{CNT} \cos\theta_i \tag{3c}$$

where L_{CNT} is the average length of the CNT; φ_i is the azimuthal angle; and θ_i is the polar angle.

The orientation of CNT_i is defined by the azimuthal and polar angles, which can be generated by Equations (4a) and (4b).

$$\varphi_i = 2\pi \cdot rand \tag{4a}$$

$$\cos\theta_i = (1 - \cos\theta_{max}) \times rand + \cos\theta_{max} \tag{4b}$$

where θ_{max} is the upper limit of the alignment angle (i.e., the angle between the CNT and the x-axis). θ_{max} equals to $\pi/2$ represents the case of PNCs with isotropically distributed CNTs without any preferential orientation; θ_{max} equals to zero represents the case of PNCs with all CNTs perfectly aligned in the direction of the current flow (i.e., the *x*-axis).

2.2 Electrical conductivity of PNCs represented by an individual cuboid RVE

In the simulation model, all the interconnecting CNTs that bridge the two electrodes in a RVE are identified to construct the random resistor network. Similar to the previous numerical studies [22, 23], two types of resistances in the percolating CNTs network are considered, namely the intrinsic resistance ($R_{intrinsic}$) along an individual CNT and the contact resistance ($R_{contact}$) between two CNTs. Referring to Figure 2, two CNTs are considered to be connected and form a continuous electrically conductive path when the shortest distance between them is shorter than the cut-off distance (d_{cutoff}). The intrinsic resistance along CNT_i, between $Node_k$ and $Node_l$, can be calculated by Equation (5) [22, 23].

$$R_{intrinsic,kl} = \frac{4l_{kl}}{\pi\sigma_{CNT}D^2} \tag{5}$$

where σ_{CNT} is the intrinsic electrical conductivity of the CNT and D is the diameter of the CNT.

The contact resistance between CNT_i and CNT_j is induced by electron ballistic tunneling through the contact junction. The Landauer-Büttiker formula can be used to estimate the contact resistance [29-33], which depends on the shortest distance between them (i.e., d_{kp}), which is shorter than d_{cutoff}. Its value can be determined by Equations (6a) through (6c).

$$R_{contact,kp} = \frac{h}{2e^2}\frac{1}{M\mathcal{T}} \tag{6a}$$

$$\mathcal{T} = \begin{cases} \exp\left(-\frac{d_{vdw}}{d_{tunnel}}\right) & \text{for } 0 \le d_{kp} \le D + d_{vdw} \\ \exp\left(-\frac{d_{kp} - D}{d_{tunnel}}\right) & \text{for } D + d_{vdw} \le d_{kp} \le D + d_{cutoff} \end{cases} \tag{6b}$$

$$d_{tunnel} = \frac{h}{2\pi} \frac{1}{\sqrt{2m_e \Delta E}} \tag{6c}$$

where h is Planck's constant; $\mathcal{T}$ is the transmission probability for the electron to tunneling between CNTs through polymer; M is the number of conduction channels, which is dimensionless and related to CNT's diameter [34]; e is the charge of an electron; d_{vdw} is the van der Waals separation distance caused by the Pauli exclusion principle [35, 36], which limits the minimum distance between two CNTs; d_{tunnel} is the tunneling characteristic length; m_e is the mass of an electron; and ΔE is the height of energy barrier [37].

After identifying the random resistor network of CNTs in the RVE as well as calculating the $R_{intrinsic}$ and $R_{contact}$ of all CNTs in the network, a conductance matrix is constructed to represent the network using the Kirchhoff's circuit laws [38]. The equivalent conductance of this conductive network (i.e., denoted as G_{RVE}) can be obtained using the Gaussian decomposition method [38]. Consequently, the PNC's electrical conductivity (σ) can be obtained by Equation (7).

$$\sigma = G_{RVE} \frac{l}{A} = G_{RVE} \frac{L_x}{L_y L_z} \tag{7}$$

where l is the distance between the left and right boundary surfaces; and A is the cross-sectional area of the RVE.

2.3 Electrical conductivity of PNCs by considering the interconnectivity of CNTs across the RVEs' boundary surfaces

Figure 3 shows a schematic that illustrates the interconnectivity of CNTs across a boundary surface between adjacent RVEs. It is apparent that the simulation procedures described in the previous sections will neglect the highlighted conductive pathways generated by CNTs across the boundary surface (i.e., path AA' and BB' in Figure (4)). This would lead to an underestimation of the PNC's σ and an overestimation of the PNC's ϕ_c by the model, especially for a relatively small RVE. One possible solution to circumvent this problem is to use a RVE that is sufficiently large such that the error caused by the ignorance of the cross-boundary CNTs would reduce. However, such approach will significantly increase the computation cost (i.e., time and memory) to run the simulation. Therefore, a new model is developed in this work to account for the electrical conductivity contributed by the interconnecting CNTs in adjacent RVEs across the boundary surfaces. Without adopting the "cut-and-relocate" approach to handle CNTs penetrating the RVE's boundary, it is possible to avoid the unnecessary bias that immediately increases the number of CNTs penetrating the two terminals. This will help filling a loophole existed in previous simulation models, leading to more realistic representation of electrically conductive PNC filled embedded with three-dimensional CNT networks. In other words, the new model will enhance the research on the structure-to-property relationships of these PNC, and thereby facilitate the development of smart and multifunctional PNC with tailored electrical properties.

In the new model, the CNTs are distributed in an $L_x \times L_y \times 2L_z$ rectangular cuboid, consisting of a RVE on top of another, as illustrated in Figure 4. The electrical conductance of the rectangular cuboid, the top RVE, and the bottom RVE (i.e., G_{rect},

$G_{RVE,top}$, and $G_{RVE,bottom}$) can be calculated independently by the procedures discussed in the previous sections. In Figure 4, the CNTs penetrating across the boundary surface are highlighted. Among them, some CNTs contributing to the electrically conductive pathways in the rectangular cuboid cell may contribute neither to the electrically conductive pathways in the top RVE nor to the bottom RVE. In other words, they are independent of the CNT networks contributed to $G_{RVE,top}$ and $G_{RVE,bottom}$. These extra electrically conductive pathways would lead to additional electrical conductance in both the bottom boundary surface of the top RVE and the top boundary surface of the bottom RVE. Considering the isotropic random distribution of CNTs, the additional electrical conductance in both the bottom boundary surface of the top RVE and the top boundary surface of the bottom RVE can be approximated to equal each other, and they can be denoted as $G_{boundary}$. These additional and independent pathways ($G_{boundary}$), bridging the left boundary surface (i.e., high voltage electrode) and the right boundary surface (i.e., low voltage electrode), are parallel to the CNT networks in the cores of the top RVE ($G_{RVE,top}$) and the bottom RVE ($G_{RVE,bottom}$). Hence, the relationship among G_{rect}, G_{RVE}, and $G_{boundary}$ (i.e., the average of $G_{RVE,top}$, and $G_{RVE,bottom}$) can be expressed by Equation (8).

$$G_{rect} = 2G_{RVE} + 2G_{boundary} \tag{8}$$

Similarly, the front and back boundary surfaces of each cuboid RVE would also contribute to additional electrical conductance of the RVE. All together, the conductance of an individual cuboid RVE, accounting for the additional electrical conductance contributed by the four boundary surfaces parallel to the direction of electric current, can be determined by Equation (9).

$$G_{adj} = G_{RVE} + 4G_{boundary} \tag{9}$$

where G_{adj} and G_{RVE} are the equivalent electrical conductance with and without considering the additional conductance contributed by the interconnecting CNTs across the boundary surfaces, respectively.

Combining Equations (8) and (9), the equivalent electrical conductance of an individual RVE can be expressed as Equation (10).

$$G_{adj} = 2G_{rect} - 3G_{RVE} \tag{10}$$

Replacing G_{RVE} by G_{adj} in Equation (7), PNC's electrical conductivity (σ_{adj}), after accounting for the contribution of the interconnecting CNTs across the boundary surfaces of adjacent RVEs to the equivalent electrical conductance, can be calculated using Equation (11).

$$\sigma_{adj} = \left(2G_{rect} - 3G_{RVE}\right)\frac{L_x}{L_y L_z} \tag{11}$$

3. Results and Discussions

Using the newly developed model, Monte Carlo simulations were conducted to investigate the effects of RVE's dimensions, PNC's thickness, and CNT's alignment on PNCs' ϕ_c and σ. Simulation results were compared with experimental data and other simulation works to verify the validity of the models. Studies were also conducted to confirm the enhancement in computational efficiency by the refined modelling scheme using smaller RVEs without compromising the accuracy of the simulation results. Table 1 summarizes the values of some key physical parameters used in this work. The value of d_{vdw} was based on the Van de Waals interaction between carbon nanotubes [35, 36]. The value of d_{cutoff} was based on the threshold distance between CNTs, over which the transmission probability would be less than 10^{-6} [23]. ΔE depends on the polymer matrix [39]. Its value was chosen according to previous numerical studies on the electrical conductivity of PNCs [23, 39]. This also ensures a fair comparison between our simulation results in this study to those based on other models reported in literatures [23, 39] in the later section (i.e., Section 3.2).

Table 1. Physical Parameters used in simulation

Physical Parameters	Value	Unit
Van der Waals separation distance (d_{vdw}) [35, 36]	3.4	Å
Cut-off distance (d_{cutoff}) [23]	1.4	nm
Energy barrier (ΔE) [23, 39]	1.0	eV

3.1 Effects of RVE's dimensions on the simulation results

Various studies have demonstrated that the simulation results of PNC's σ using the 3D resistor network model are sensitive to the dimensions of the RVE. In order to suppress the effect of RVE's size on the simulation results, Hu *et al.* [22] showed that a cuboid RVE of 25 μm × 25 μm × 25 μm was needed to simulate ϕ_c and σ of PNCs filled with CNTs with an average length of 5 μm in order to achieve numerical convergence. However, the relatively large RVE also led to a surge in computational burden. In order to demonstrate the sensitivity of the simulation results based on the refined model on the RVE's dimensions, a series of simulations using RVEs with different sizes were conducted and the results are shown in Figures 5(a) through 5(d). As a case example, the physical parameters of CNTs used in the simulations were the same as those used by Hu *et al.*. Hence, the length and diameter of CNTs were set to be 5 μm and 50 nm, respectively. In addition, the number of conduction channels (M) and intrinsic electrical conductivity of a CNT (σ_{CNT}), were chosen to be 460 and 1×10^4 S/m, respectively [34].

Figure 5(a) depicts that the simulation results of PNCs' σ_{adj} were sensitive to the changes in cubic RVEs' dimensions at low CNT loadings (i.e., $\phi < 0.01$). It is apparent that simulated σ_{adj} at different CNT loadings converged as the RVEs' dimensions increased. Regardless of the RVEs' sizes, the PNCs' σ_{adj} rapidly increased at low CNT loadings and approached to a plateau as ϕ increased. The general trends were consistent with the numerical works conducted by Hu *et al.* [22]. Figures 5(b) and 5(c) plot the effect of simultaneously varying L_y and L_z while fixing L_x, and that of varying L_x while fixing L_y and L_z, respectively, on PNCs' σ_{adj}. These results would help to decouple the influence of varying the dimension parallel to the direction of current flow from that of varying the dimensions perpendicular to the current flow. As shown in Figure 5(b), when L_x was fixed to $1.5 \times L_{CNT}$, PNCs' σ_{adj} were unaffected by varying L_y and L_z, regardless of the CNT loadings. In contrast, Figure 5(c) reveals that when L_y and L_z were fixed, the simulated PNCs' σ_{adj} were sensitive to the change of L_x at low CNT loadings (i.e., $\phi < 0.01$), while they were virtually indifferent at higher CNT loadings. The figure suggested that the simulation results started to converge when L_x was greater than or equal to $4.0 \times L_{CNT}$. Figure 5(d) shows the relative difference between the simulation results with and without considering the contribution of interconnecting CNTs across the boundary surfaces (i.e., σ_{adj} and σ) when different L_y and L_z were used. The relative difference reduced as L_y and L_z increased. In other word, the contribution made by the interconnecting CNTs across the boundary surfaces to PNC's σ was more significant at smaller L_y and L_z.

Since varying L_y and L_z while fixing L_x were proven to have negligible influence on the simulation of PNCs' σ, it is possible to significantly improve the computational efficiency of the Monte Carlo simulation by reducing L_y and L_z. Figure 6 compares the simulation time per run for the case using RVE with all dimensions equaled to 4.0 × L_{CNT} with that for the case using RVE of L_x equaled to 4.0 × L_{CNT} and L_y and L_z equal to 1.1 × L_{CNT}. While the simulation results were virtually smilar in both cases, the computation time was found to be significantly shortened by reducing L_y and L_z from 4.0 × L_{CNT} to 1.1 × L_{CNT}, especially at higher CNT loadings. Therefore, the refined modelling scheme, which uses a relatively small cuboid RVE, would substantially improve the simulation efficiency without compromising the desired numerical accuracy.

3.2 Model validation and comparison with experimental works and other models

In order to verify the validity of the new model developed in this work, simulation results of σ_{adj} for PNCs filled with CNTs of different grades (i.e., different dimensions and physical properties) were compared to experimental results [26-28]. Figure 7(a) plots the comparison of simulated PNCs' σ_{adj} with experimental measurements for PNCs filled with multi-walled carbon nanotubes (MWNTs) of L_{CNT} and D equaled to 5 μm and 50 nm, respectively [26, 27]. Unlike most of other numerical studies that selected values of CNT's σ_{CNT} within the typical range reported in literature, the simulation in this work considered the dependence of σ_{CNT} and M on the dimensions of CNTs [34] in order to conduct more realistic simulation. These two parameters were determined to be 2.1 × 10^3 S/m and 460, respectively. It can be observed that the numerical predictions of PNCs' σ_{adj} at different CNT loadings showed good agreement with experimental results. Figure 7(b) illustrates the comparison of simulated PNCs' σ_{adj} with experimental data for PNCs filled with MWNTs of smaller dimensions yet higher aspect ratio. Distribution of L_{CNT} was characterized by Krause *et al*. using a normalized frequency versus CNT's length bar chart [38]. In our simulation work, the

Weibull Distribution [40], as Equation (12), was used to represent such distribution of L_{CNT}.

$$f(x;\lambda,k)=\begin{cases}\dfrac{k}{\lambda}\left(\dfrac{x}{\lambda}\right)^{k-1}e^{-\left(\frac{x}{\lambda}\right)^{k}} & x\geq 0,\\ 0 & x<0,\end{cases} \tag{12}$$

where λ is the scale parameter and k is the shape parameter.

The average length and diameter of the MWNTs were determined to be 1.603 μm and 9.5 nm, respectively. The resultant σ_{CNT} and M were equal to 900×10^3 S/m and 87, respectively [34]. Similar to the previous case, the PNCs' σ_{adj}, numerically predicted by the new model and simulation scheme, showed good agreement with the experimental measurements. In short, the good agreement between the simulation results and the experimental results for the electrical conductivity of PNCs filled with different grades of MWNTs suggest that the refined model and simulation scheme can be used as a strategic tool to design CNT loadings and CNTs' degree of alignment to tailor the PNCs' ϕ_c and σ. This will facilitate the identification of structure-to-property relationship of electrically conductive PNC embedded with three-dimensional CNT network. Moreover, it will enhance the research and development of smart and multifunctional PNC with tailored electrical properties.

Simulations results obtained using the refined model were also compared with other numerical studies, based on the 3D random resistor network model, conducted by Hu *et al.* [22] and Bao *et al.* [23]. The comparisons are plotted in Figures 8(a) and 8(b). Since Hu *et al.* employed a cubic RVE with side length of $5 \times L_{CNT}$, the RVE used herein had L_x equaled to $5 \times L_{CNT}$, while L_y and L_z equaled to $1.1 \times L_{CNT}$ in order to ensure a fair comparison. The reduced lengths in both L_y and L_z were justified by the insensitivity of the simulation results to the changes in these two dimensions. Bao *et al*. [23], on the other hand, employed a cubic RVE of its side length equal to 1.1 × L_{CNT}. Therefore, to ensure a fair comparison, the simulation results to be compared with their results were generated based on two cases: (i) a RVE of identical dimensions $L_x = L_y = L_z = 1.1 \times L_{CNT}$; and (ii) a RVE of L_x equaled to 4 × L_{CNT}, while L_y and L_z equaled to 1.1 × L_{CNT}.

Figure 8(a) illustrates the comparison between the simulation results of σ and σ_{adj} and the predicted σ obtained by Hu *et al.* [22]. The plot reveals that σ_{adj} (i.e., predicted σ by considering interconnecting CNTs at the boundary) was higher than σ (i.e., predicted σ without considering interconnecting CNTs at the boundary) for PNCs with different CNT loadings; however, the difference between them reduced with increasing loads of CNT. This trend can be attributed to the establishment of sufficient number of electrically conductive paths in the PNC's core at higher CNT contents. In other words, the relative contribution of interconnecting CNTs across the RVE's boundary surface to the overall conductive network gradually decreased. Moreover, Figure 8(a) also reveals that the simulation results reported by Hu *et al.* were consistent with σ_{adj} and σ simulated in this work at a wide range of CNT loadings. The similar values of ϕ_c are obtained. Nevertheless, obvious differences can be observed for PNCs with CNT's volume fraction either below ~0.013 or above ~0.03. At lower CNT loading (i.e., $\phi <$ 0.013), by cutting the extruding portion of CNTs outside the RVE and relocating them onto the opposite boundary surfaces, Hu *et al.* unintentionally generated bias on the random distribution of CNTs and resulted in a higher than usual number of CNTs in contact with the two electrodes (i.e., the left and right boundaries of RVE). This led to a higher simulated σ than our predicted σ. At higher CNT loading (i.e., $\phi >$ 0.03), the error caused by the ignorance of the contact resistance among interconnecting CNTs by their work became more significant as the number of interconnecting CNTs in the random resistor network increased. Therefore, it would lead to an overestimation of PNCs' σ, which resulted in a simulation result higher than the predicted σ in this work. This was also consistent with the work by Hu *et al.* [39] that demonstrated the significance of tunneling effect on PNCs' σ.

Figure 8(b) illustrates the comparison between the simulation results by Bao *et al.* [23] and the simulated σ_{adj} based on RVEs of the aforementioned two dimensions. Comparing the trends among the three plots, it is worthwhile to note that the trends of our calculated σ_{adj} using a cubic RVE with its side length equaled to $1.1 \times L_{CNT}$ was similar to the simulated σ reported by Bao *et al.* This also resulted in very similar ϕ_c between their work (i.e., $\phi_c = 0.00279$) and this study (i.e., $\phi_c = 0.00288$). However, it is apparent that their predicted σ was consistently lower than the simulated σ_{adj} in this work at all CNT loadings. Using the "cut and relocate" approach, the CNT's length, and thereby the aspect ratio in their relatively small RVE would be decreased significantly. This would reduce the probability to generate an electrically conductive pathway, leading to lower predicted values of PNC's σ. Moreover, obvious difference can be seen when comparing their predicted σ with the simulated σ_{adj} based on a RVE with L_x equaled to $4 \times L_{CNT}$. By increasing L_x to $4 \times L_{CNT}$, the ϕ_c of the PNC increased dramatically (i.e., $\phi_c = 0.00546$) because of the lower probability to develop a continuous conductive channel along a longer path. This again confirmed that a sufficient length in the direction of current flow is needed to achieve numerical convergence.

3.3 Effect of the distance between two electrodes in a RVE on percolation threshold

As previously shown in Figure 5(c), the simulated PNC's σ_{adj} was predominantly influenced by the change in L_x (i.e., the distance between two electrodes in a RVE), especially at low CNT loadings (i.e., $\phi < 0.01$). Such influence was negligible at higher CNT loadings. To more clearly demonstrate the effect of changing L_x on the formation of conductive CNT network in a polymer matrix, the power law fitting based on the percolation theory (i.e., Equation (1)) was applied to determine the PNC's ϕ_c for varying L_x. It is apparent that ϕ_c increased as the volume of the RVE increased by only increasing L_x. In contrast, when the volume of the RVE increased by simultaneously increasing both L_y and L_z, the PNC's ϕ_c remained relatively constant. Overall, it is reasonable to conclude that ϕ_c has a positive relationship with L_x, but is virtually independent of L_x and L_z. This finding on the effect of increasing L_x on ϕ_c is consistent with the experimental observations reported by Fu *et al.* [25]. It was proven experimentally that PNC's ϕ_c would be influenced by the test sample's thickness, across which the PNC's σ was measured. Their experimental measurements revealed that PNC's σ increased significantly with the thickness of the PNC sample.

The positive relationship between L_x and PNC's ϕ_c can be explained by considering the probability of having the minimum number of CNTs to construct an electrically conductive pathway across the two terminals. Under the percolation theory, ϕ_c is the critical concentration of CNTs at which the formation of the first conductive CNT pathway, bridging the two electrodes across the RVE (i.e., across $L_{x,}$). Increasing $L_{x,}$means a longer separation distance between the two electrodes, leading to the needs of a larger number of CNTs to construct a conductive path. For a given loading of randomly dispersed CNTs in the RVE, the probability of sequentially interconnecting

CNTs to form a continuous path to bridge the two electrodes decreases exponentially as the minimum required number of CNTs increases. For example, for a RVE with L_x equals to 10 μm, it requires at least two 5 μm-long CNTs to form a conductive path bridging the two electrode surfaces. If L_x of the RVE is increased to 15 μm, a minimum of three 5 μm-long CNTs would be needed to achieve the same. The probability of three CNTs to interconnecting one another can be determined by considering the overall probability of two independent events to occur simultaneously. First, two CNTs need to be interconnected to create a conductive channel. Consequently, a third CNT will need to interconnect with the end of this two-CNT channel. It is apparent that the probability of forming a three-CNT conductive pathway (i.e., consists of two independent events) is lower than that of forming a two-CNT conductive pathway. Therefore, it would be less likely for a small number of CNTs to generate a conductive pathway across the RVE with longer L_x. In other words, a higher concentration of CNTs would be required to construct an electrically conductive pathway, and thereby increasing the PNC's ϕ_c. On the other hand, increasing L_y and L_z only lead to the increase in the RVE's volume without affecting the probability of developing the first conductive path in the direction of the electric current flow.

3.4 Effect of CNTs' alignment on PNCs' percolation thresholds and electrical conductivity

Figures 10(a) through 10(c) plot the effect of CNTs' alignment on the PNCs' ϕ_c and σ_{adj}. The simulations were conducted using a RVE as described in Section 3.1. The alignment of CNTs was varied by setting the upper limits for the angle between CNTs and the *x*-axis (i.e., θ_{max}) in Equation (4b). Figure 10(a) shows that the simulation results of PNCs' σ_{adj} versus the degree of CNT alignment. For θ_{max} ranging for 30° (i.e., highly aligned) to 90° (i.e., perfectly random orientation), the PNC's σ_{adj} increased rapidly at low CNT loading, and increased gradually thereafter to a plateau with increasing CNT loadings. However, the dependence of PNC's σ_{adj} on θ_{max} varied with CNT loadings. This changing relationship of σ_{adj} versus θ_{max} is clearly revealed in Figure 10(b). In order to maximize PNC's σ_{adj}, a higher degree of CNT alignment was favorable at higher CNT loadings, whereas more randomly oriented CNTs were beneficial at lower CNT loadings. Furthermore, the effects of CNTs alignment on PNCs' ϕ_c are shown in Figure 10(c). The simulation results show that ϕ_c decreased as θ_{max} increased (i.e., CNTs were more randomly oriented). These findings were consistent with the experimental studies conducted by Du *et al.* [18].

When CNTs are highly aligned, (e.g., θ_{max} = 30°), all CNTs are nearly parallel to each other. Thus, it is less probable for CNTs to interconnecting each other to form a conductive channel at small ϕ. In other words, a higher loading of CNTs would be required to form the first electrically conductive pathway, and thereby increasing PNCs' ϕ_c. On the other hand, for PNCs filled with high concentrations of CNTs, a large number of electrically conductive pathways would be formed easily in the polymer matrix. Along each of these numerous pathways, the conductance is governed by the length of the path (i.e., intrinsic resistance) and the number of CNT intersections (i.e., contact resistance) along the path. In PNCs filled with randomly-oriented CNTs, these conductive paths comprise of more CNTs, and thereby leading to longer pathways and more CNT-to-CNT intersections along these paths. As a result, the conductance of these paths would be lower, resulting in reduced PNC's σ_{adj} at higher CNT loadings. Combining the observations from Figures 10(a) through 10(c), it can be concluded that highly aligned CNTs would help to maximize PNCs' σ with high CNT loadings; however, perfectly oriented CNTs would help to suppress PNCs' ϕ_c.

4. Conclusion

A more realistic and efficient model to describe the three-dimensional carbon nanotube (CNT) networks in polymer nancomposites (PNC) has been developed in this work. Compared to various existing random resistor models [19, 20], the refined model accounts for the contribution of interconnecting CNTs across the boundary surfaces of the representative volume element (RVE). This avoid unnecessary biases in the spatial dispersion and length distribution of CNTs being caused by the "cut-and-relocate" approach adopted in typical models found in literatures. A series of Monte Carlo simulations, based on RVEs with different dimensions, have revealed that the simulated PNC's electrical conductivity (σ) is only influenced by the RVE's dimension parallel to the direction of the current flow. This leads to a more efficient simulation scheme of using RVEs with reduced length in dimensions perpendicular to the direction of the current flow. As a result, the computation cost of the Monte Carlo simulations can be suppressed significantly without compromising their accuracy. Using the improved model, σ of PNCs filled with different grades of multi-walled carbon nanotubes were predicted, and the results demonstrated good agreement with those reported in existing experimental studies. It has also been shown that PNCs' percolation threshold (ϕ_c) increased with the distance between the two electrodes of the RVE. Through comparisons with other simulation models, the key differences and improvements of the refined model were elucidated. Furthermore, the effects of CNTs' orientation on PNCs' ϕ_c and σ have been studied. Simulation results suggest that perfectly random orientation of CNTs lead to lower ϕ_c, while highly aligned CNTs along the direction of electric current lead to higher σ at high CNT loadings. In short, the new model being developed in this work would serve as a more realistic strategic tool to design optimal CNT loading and orientation to tailor the PNC's electric properties.

5. Acknowledgement

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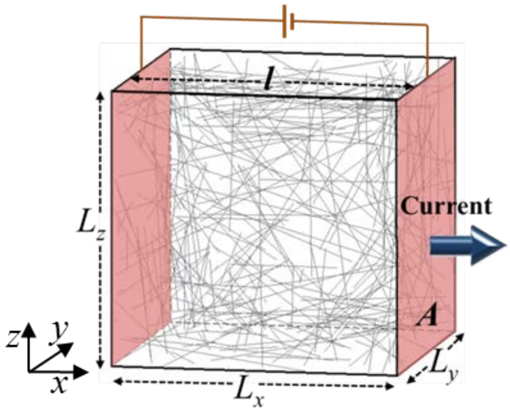


Figure 1. A schematic of CNTs randomly dispersed in a representative volume element

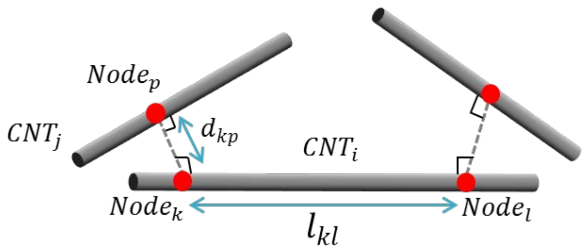


Figure 2. A schematic of CNTs' interconnection.

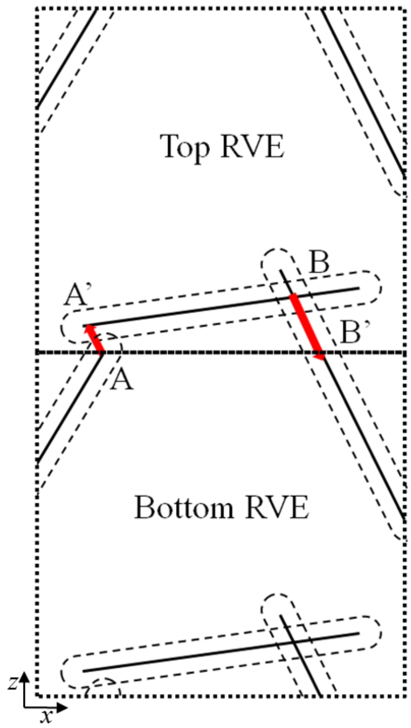


Figure 3. A schematic of interconnectivity of CNTs across the boundary surface of adjacent RVEs.

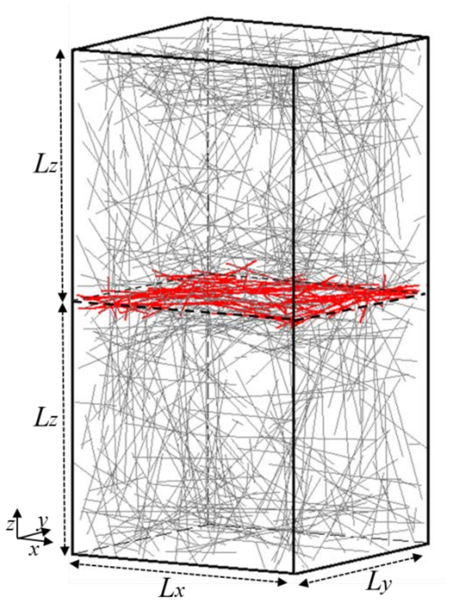
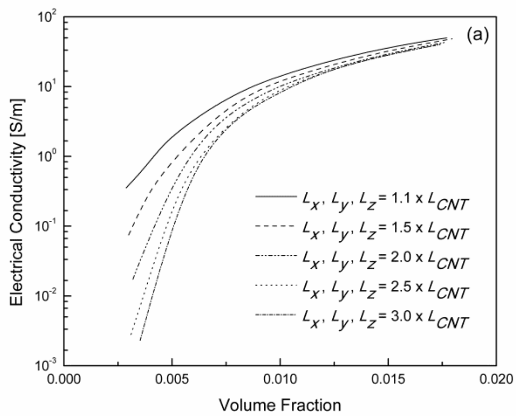


Figure 4. CNTs distributed in a rectangular cuboid (note: CNTs penetrating across the boundary surface are highlighted).



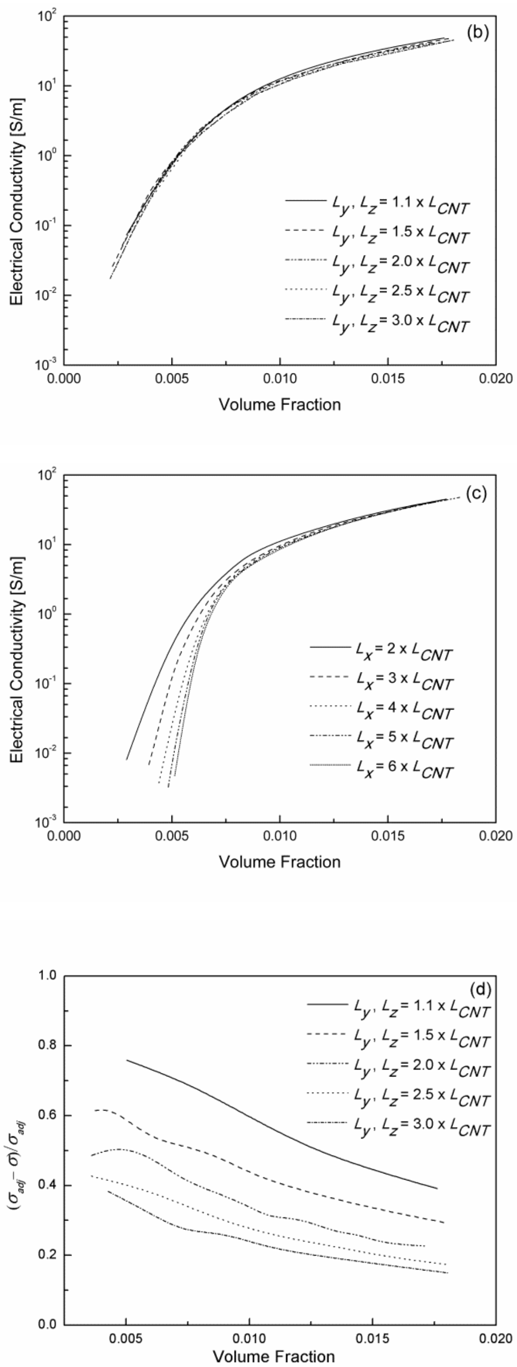


Figure 5. Simulation results of PNC's σ_{adj} at different CNT loadings based on (a) cubic RVEs with different dimensions; (b) cuboid RVEs with fixed L_x (i.e., 1.5 × L_{CNT}) but varying L_y and L_z; and (c) cuboid RVEs with fixed L_y and L_z (i.e., 1.5 × L_{CNT}) but varying L_x; (d) relative difference of simulated σ_{adj} and σ with varying L_y and L_z.

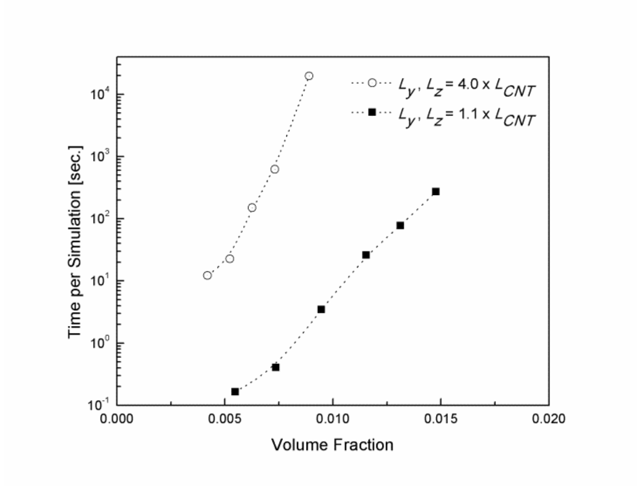


Figure 6. Comparison of simulation times per run using different lengths of L_y and L_z (note: L_x is fixed at 4 × L_{CNT}).

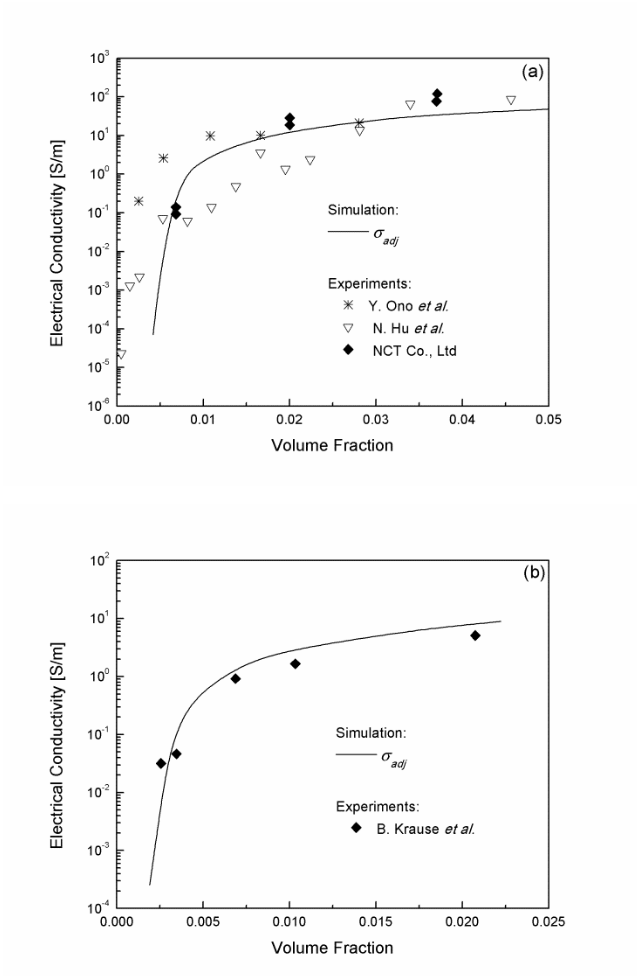
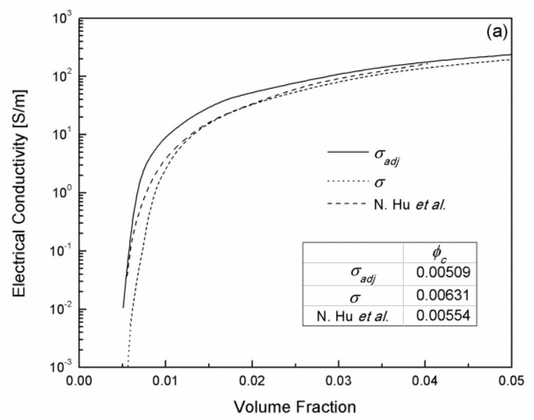


Figure 7. Comparison of simulation results of PNCs' σ_{adj} with existing experimental data obtained by: (a) Y. Ono *et al.*, N. Hu *et al.*, and NCT Co. Ltd. [26, 27]; and (b) B. Krause *et al.* [28].



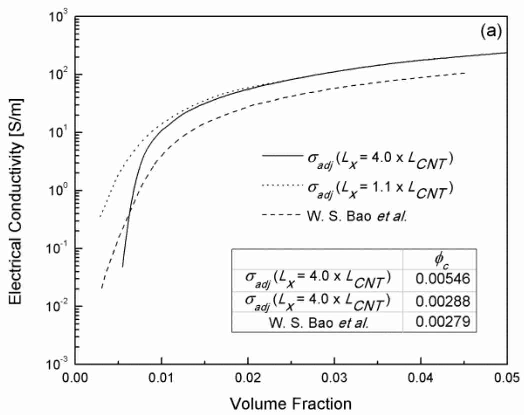


Figure 8. Comparison of simulation results with other numerical studies [22, 23] based on the 3D resistor network model.

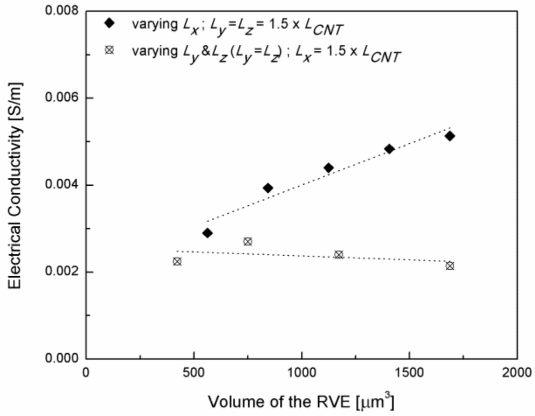
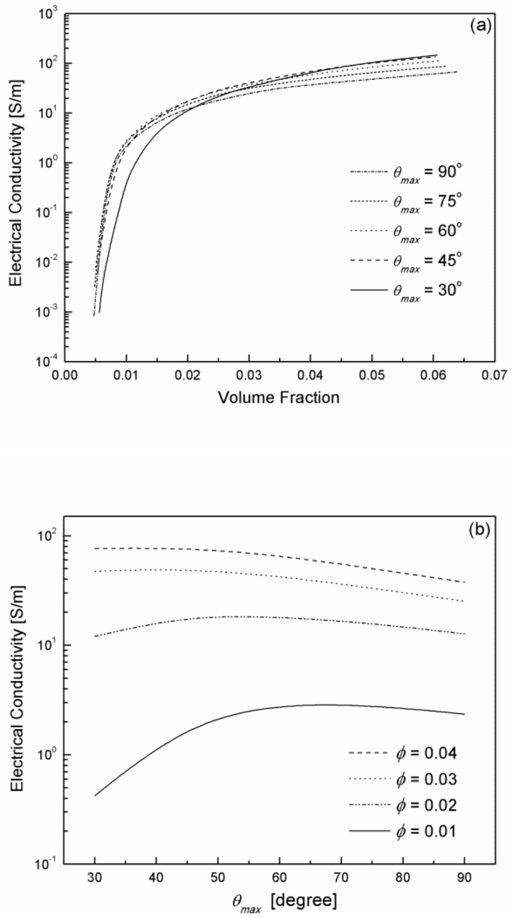


Figure 9. Calculated percolation threshold using different RVEs, cuboid RVEs with fixed L_y and L_z (i.e., 1.5 × L_{CNT}) but varying L_x; and cuboid RVEs with fixed L_x (i.e., 1.5 × L_{CNT}) but varying L_y and L_z.



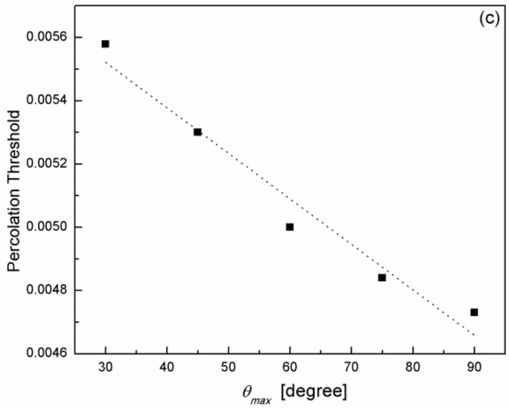


Figure 10. Calculated electrical properties of PNCs at different CNTs alignment, (a) electrical conductivity vs. volume fraction at different θ_{max}; (b) electrical conductivity vs. θ_{max} at different volume fraction; (c) percolation threshold vs. θ_{max}.